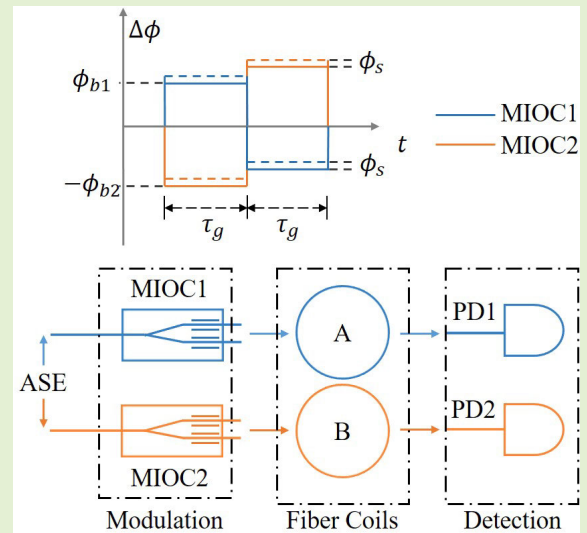


Cooperative Four-State Modulation for Sensitivity Enhancement in a Fiber Optic Gyroscope

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Xinyu Cao, Fangshuo Shi, and Zhengbin Li^{ID}

Abstract—A highly sensitive portable fiber optic gyroscope has significant value in inertial navigation and geophysical exploration. However, the excessive relative intensity noise (RIN) generated by broadband light sources degrades the sensitivity of the interferometric fiber optic gyroscope (IFOG). To suppress RIN, we propose the cooperative four-state modulation scheme based on a source-sharing configuration. Reverse-phase square-wave modulation signals are applied to two minimum configurations. The modulation states in the configuration are cooperatively implemented and therefore have RIN compensation ability. In this study, a noise model is employed to guide the selection of modulation depths. Compared to the minimum configuration with conventional sinusoidal and square-wave modulation, the new scheme achieves a 20.0-fold and 4.7-fold decrease in self-noise, as well as an 18.0-fold and 4.3-fold decrease in angular random walk. In a test on detecting the rotation rate of the Earth, a self-noise level of 55 nrad/s/Hz^{1/2} and an angular random walk of 1.5×10^{-4} deg/h^{1/2} are achieved using the proposed scheme with an enclosed area of only 25.5 m², demonstrating superior sensitivity performance.

Index Terms—Cooperative four-state modulation, interferometric fiber optic gyroscope (IFOG), modulation depth, sensitivity enhancement, source-sharing configuration.



I. INTRODUCTION

THE interferometric fiber optic gyroscope (IFOG) is a kind of sensor that measures the rotational speed of an object

based on the Sagnac effect [1]. It is widely utilized in various fields such as inertial navigation [2], [3], rotational seismology [4], [5], and gravity gradient detection [6], owing to its compact structure, all-solid-state design, and high reliability [1], [7]. In all application domains, sensitivity is a critical performance metric. Enhancing the sensitivity of IFOG is an important research topic. The sensitivity of an IFOG is significantly affected by self-noise. The primary noise components in an IFOG include relative intensity noise (RIN), thermal phase noise (TPN), Johnson–Nyquist noise, and shot noise [6], [8]. Among all the noise components, RIN constitutes the prominent component [9], [10], [11]. It originates from the random beat effect between different frequency components of the broadband light source [12], [13]. This beat effect leads to random fluctuations in the optical power of the light source.

Previous studies have primarily focused on suppressing RIN by modifying the spectral characteristics of the light source or introducing optical reference arms. An RIN-reduced IFOG

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light source using an erbium-doped fiber amplifier (EDFA) in concern with a semiconductor optical amplifier (SOA) was presented [14]. The SOA in the deep saturated region acts as a filter suppressing the fluctuations. A fiber ring resonator was used to filter the spectrum of a broadband light source at the Eigen frequency [15]. However, the RIN suppression effect achieved through such schemes is limited. The optical reference arm refers to a separate path being extracted from the light source section, which is used to monitor the fluctuation of the output intensity of the light source. This serves to compensate for the RIN, leading to the derivation of various demodulation algorithms. Typical methods are intensity noise subtraction [16] or division [17]. A more convenient approach is to add reference signal and measured signal optically to lower the excess RIN at Eigen frequency [18]. However, inconsistencies between the reference and measured signals [19] are inevitable, such as deviation of the time difference. These imperfections might degrade the effectiveness of RIN suppression.

In this article, we introduce the cooperative four-state modulation scheme based on a source-sharing configuration, aiming to suppress RIN. The overall system avoids introducing external feedback control, thereby preventing the sensitivity reduction caused by deadzone issue [20], [21]. Additionally, the advantage of high signal-to-noise ratio (SNR) in square-wave modulation [22] is taken. The optical configuration utilizes two minimum configurations that multiplex a broadband source. Two polarization maintaining fiber (PMF) coils with identical parameters are wound around the same base. Reverse-phase square-wave signals are used to modulate the two minimum configurations. In the concurrently implemented modulation states, there exist a significant amount of correlated RIN components. Therefore, the modulation state can be a reference to each other. With the optical structure and modulation scheme, RIN can be effectively suppressed while the demodulation stability is guaranteed [23]. The scheme avoids introducing additional optical reference arm, thus preventing inconsistencies between the reference and measured signals. Compared to the minimum configuration using conventional sinusoidal and square-wave modulation, the scheme achieves a 20.0-fold and 4.7-fold self-noise reduction, and an 18.0-fold and 4.3-fold decrease in angular random walk. Achieving a self-noise of 55 nrad/s/Hz^{1/2} and an angular random walk of 1.5×10^{-4} deg/h^{1/2} with an enclosed area of only 25.5 m², the new scheme demonstrates superior sensitivity.

This article is organized as follows. In Section II, we analyze the four types of noise in IFOG, including their generation mechanisms and impact. In Section III, we provide detailed information on the source-sharing optical configuration and the cooperative four-state modulation process. Further, the RIN suppression theory based on this scheme is proposed. In Section IV, we establish a numerical simulation model based on the noise mechanisms described in Section II. The simulated results align well with experimental results of conventional square-wave modulation. Next, the model is utilized to simulate the cooperative four-state modulation process, which provides a basis for modulation depth selection and predicts the RIN suppression capability of the scheme.

In Section V, we verify the effectiveness of the scheme through experiments and compare it to the commonly used sinusoidal modulation and square-wave modulation.

II. NOISE MECHANISMS OF IFOG

The intrinsic noise impacting the sensitivity of IFOG primarily encompasses RIN, TPN, Johnson–Nyquist noise, and shot noise. To enhance the sensitivity, it is necessary to decompose the noise and propose targeted suppression techniques.

The use of a broadband light source is necessary because of its short coherence length [24]. RIN is the power fluctuation caused by random beats between uncorrelated frequency components in the broadband source spectrum. For IFOG with amplified spontaneous emission (ASE) light source, the light intensity correlation function $\Gamma_I(\tau)$ can be expressed as the fourth-order coherence function of the electric field $E(t)$ [25]. For IFOG utilizing all-polarization-maintaining components, the transmitted light can be approximated as fully polarized light. Therefore, $\Gamma_I(\tau)$ is represented as the second-order correlation function of the intensity function I [25]

$$\begin{aligned}\Gamma_I(\tau) &= \langle E(t)E^*(t)E(t+\tau)E^*(t+\tau) \rangle \\ &= I^2[1 + |\gamma(\tau)|^2]\end{aligned}\quad (1)$$

where $\langle \rangle$ represents time-domain averaging and $\gamma(\tau)$ is the complex coherence degree of light. The power spectral density (PSD) $S_{\text{RIN}}(f)$ of RIN is further expressed as

$$S_{\text{RIN}}(f) = \mathcal{F}[\Gamma_I(\tau)] = I^2 \int_0^{+\infty} p(v)p(v+f)dv \quad (2)$$

where $p(v)$ is the normalized PSD function. The frequency bandwidth of the wide-spectrum light source is about 2 THz. However, limited by photoelectric converters and A/D conversion devices, the detection frequency bandwidth of the interference signal is several megahertz, which is far less than the spectral width of the light source. So $S_{\text{RIN}}(f) \approx S_{\text{RIN}}(0)$. In particular, when using ASE light source with a spectrum of rectangular form, $S_{\text{RIN}}(f)$ is simply demonstrated as the reciprocal of the source bandwidth Δf_s , that is,

$$S_{\text{RIN}}(f) = \frac{I^2}{\Delta f_s}. \quad (3)$$

The temperature-induced fluctuations in the refractive index of the fiber lead to phase variations of transmitted light. This results in the emergence of phase fluctuations superimposed on the Sagnac phase. The noise is referred to as TPN, and its spectral density $S_{\text{TPN}}(f)$ can be approximated and expressed as follows [26]:

$$\begin{aligned}S_{\text{TPN}}(f) &= \sqrt{\frac{k_B T^2 L}{\kappa \lambda^2} \left(\frac{dn_{\text{eff}}}{dT} + n_{\text{eff}} \alpha \right)^2} \\ &\times \sqrt{\ln \left[\frac{\left(\frac{2}{W_0} \right)^4 + \left(\frac{2\pi f}{\xi} \right)^2}{\left(\frac{4.81}{d} \right)^4 + \left(\frac{2\pi f}{\xi} \right)^2} \right]} \times \left[1 - \text{sinc} \left(\frac{2\pi f}{v} \right) \right]\end{aligned}\quad (4)$$

where k_B is the Boltzmann's constant, T is the absolute temperature in Kelvin. L is the total length of the fiber.

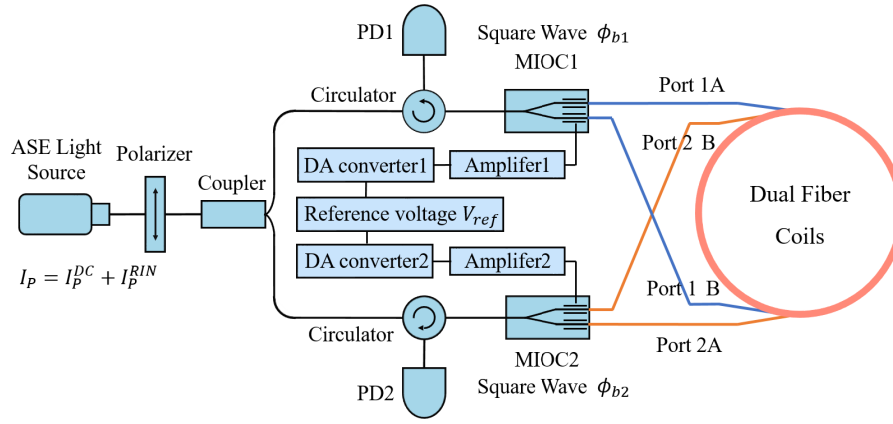


Fig. 1. Optical configuration of cooperative four-state modulation. Two minimum configurations share the same ASE light source. Two D/A converters share a reference voltage. Square waves with modulation depths of ϕ_{b1} and ϕ_{b2} are, respectively, applied to MIOC1 and MIOC2.

n_{eff} represents effective refractive index. κ is the thermal conductivity. α is the linear thermal expansion coefficient, and dn_{eff}/dT is the temperature coefficient of the effective refractive index in the fiber. Furthermore, W_0 and d are the mode field radius and the cladding diameter of the fiber, ξ and v are the thermal diffusivity and effective speed of light in the fiber.

Johnson–Nyquist noise is generated by fluctuations in the carrier wave within the trans-impedance amplifier of a detector. These fluctuations appear as power variations in the dissipative circuit elements. The independence of carrier motion allows Johnson–Nyquist noise to be modeled as a stationary Gaussian random process with a zero mean [27], in accordance with the central limit theorem. The electric current standard deviation N_{J-N} is expressed as [28]

$$N_{J-N} = \sqrt{\frac{4k_B T \Delta f}{R}} \quad (5)$$

where R is detector resistance and Δf is the measurement bandwidth.

Shot noise arises from the discrete flows of charge carriers in detectors, and sets the quantum limit for passive optical systems. The electric current standard deviation N_{shot} is expressed as [28]

$$N_{\text{shot}} = \sqrt{2e\Delta f \eta I} \quad (6)$$

where e represents the charge of an electron and η is the responsivity of the photodetector.

The intrinsic noise power of an IFOG is the summation of the aforementioned components and RIN is a major contributor to total noise [29]. Increasing the sampling rate of the optical signal or the bandwidth of light source are direct ways to enhance sensitivity. However, these methods increase the cost of sensors and have limited effectiveness. It is worth noting that decreasing the average light intensity can suppress RIN, but the influence of Johnson–Nyquist noise and shot noise will increase. Implementing effective RIN suppression methods, while reducing the impact of other noises, can significantly improve the sensitivity of IFOG.

III. COOPERATIVE FOUR-STATE MODULATION

A. Optical Path and Modulation Scheme

The cooperative four-state modulation scheme constructed by using the source-sharing optical configuration is illustrated in Fig. 1. The light propagation process begins with an ASE light source. After passing through a polarizer, the light becomes linearly polarized and is then evenly split into two beams with identical power using a 50/50 polarization maintenance coupler. The polarizer provides the polarization reference, which is the basis of RIN suppression. Two multifunctional integrated optical chips (MIOCs) are used to perform beam splitting and modulation of polarized light. Square waves with modulation depths of ϕ_{b1} and ϕ_{b2} are, respectively, applied to MIOC1 and MIOC2. The phase difference between the two signals remains at π rad. Two PMF coils are wound on the same base with quadrupolar winding pattern. Based on the Sagnac effect, the counter-propagating light waves experience a phase difference proportional to the rotation speed. The light in PMF coil A and PMF coil B is captured by photodetector 1 (PD1) and photodetector 2 (PD2), respectively.

For the source-sharing configuration with cooperative four-state modulation, the two electric field intensities after coupler are represented as follows: $\mathbf{E}_A = E(t)e^{j\omega t}$, $\mathbf{E}_B = E(t)e^{-j\phi_c}e^{j\omega t}$. $E(t)$ represents amplitude of electric field and ω is the center frequency of the ASE light source. ϕ_c is the phase shift caused by coupler. The intensity fluctuations in the two PMF coils are consistent, which provides assurance for suppressing RIN [9].

The modulation model is illustrated in Fig. 2. The two modulation signals are applied to MIOC1 and MIOC2, respectively. The modulation frequency $f_m = 1/2\tau_g$, where τ_g represents the traveling time of light in a PMF coil. Traveling time $\tau_g = n_{\text{eff}}L/c$, where c represents the speed of light in vacuum. The minimum configuration containing PMF coil A operates at bias points of ϕ_{b1} and $-\phi_{b1}$ during two half periods with a duration of τ_g . The two states are denoted as I_{A1} and I_{A2} . The minimum configuration containing PMF coil B operates at bias points of $-\phi_{b2}$ and ϕ_{b2} during two half periods with a duration of τ_g . The two states are denoted as I_{B1}

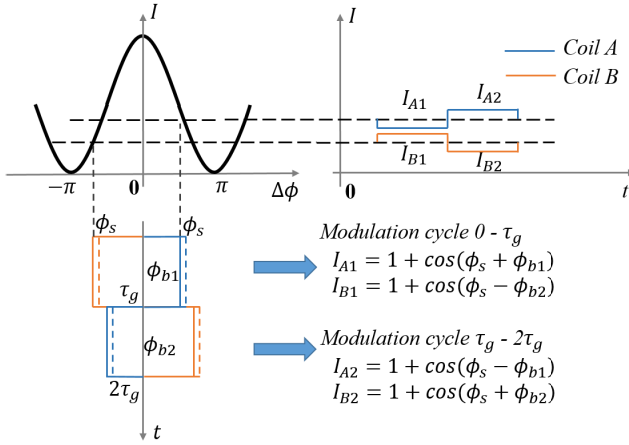


Fig. 2. Schematic of cooperative four-state modulation. The phase difference between the two square waves remains at π rad. The two states of light in coil A are denoted as I_{A1} and I_{A2} . The two states of light in coil B are denoted as I_{B1} and I_{B2} . I_{A1} and I_{B1} are implemented concurrently. I_{A2} and I_{B2} are implemented concurrently.

and I_{B2} . The light intensities of the four states are given by (7). The Sagnac phase shift ϕ_s is solved from (8)

$$I_{A1} = I_p + I_p \cos(\phi_s + \phi_{b1}) \quad (7a)$$

$$I_{A2} = I_p + I_p \cos(\phi_s - \phi_{b1}) \quad (7b)$$

$$I_{B1} = I_p + I_p \cos(\phi_s - \phi_{b2}) \quad (7c)$$

$$I_{B2} = I_p + I_p \cos(\phi_s + \phi_{b2}) \quad (7d)$$

$$\phi_s = \arctan \left[\frac{v_{cf} - 1}{v_{cf} + 1} \tan \frac{(\phi_{b1} - \phi_{b2})}{2} \right] \quad (8)$$

$$v_{cf} = \frac{I_{A1} - I_{B1}}{I_{A2} - I_{B2}}.$$

B. Theoretical Model of RIN Suppression

As described in Section II, the random beat between uncorrelated frequency components in the spectrum of the ASE source leads to power fluctuations. Therefore, the intensity can be expressed as $I_p = I_p^{dc} + I_p^{RIN}$, where I_p^{dc} is nontime-varying constant. In addition, I_p^{RIN} is a random function of time, representing the fluctuating component. In the modulation scheme shown in Fig. 2, the RIN of the four states can be modeled as follows:

$$I_{A1}^{RIN} = I_p^{RIN1} [1 + \cos(\phi_s + \phi_{b1})] \quad (9a)$$

$$I_{A2}^{RIN} = I_p^{RIN2} [1 + \cos(\phi_s - \phi_{b1})] \quad (9b)$$

$$I_{B1}^{RIN} = I_p^{RIN1} [1 + \cos(\phi_s - \phi_{b2})] \quad (9c)$$

$$I_{B2}^{RIN} = I_p^{RIN2} [1 + \cos(\phi_s + \phi_{b2})]. \quad (9d)$$

In (9), I_p^{RIN1} and I_p^{RIN2} are denoted as the random terms in the different halves of a modulation period. The RIN of the states implemented at the same time is highly correlated. Under the demodulation process given in (8), the excess RIN contained in the numerator and denominator of v_{cf} is represented as follows:

$$I_{A1}^{RIN} - I_{B1}^{RIN} = I_p^{RIN1} \sin \phi_s (\sin \phi_{b1} - \sin \phi_{b2}) \quad (10a)$$

$$I_{A2}^{RIN} - I_{B2}^{RIN} = -I_p^{RIN2} \sin \phi_s (\sin \phi_{b1} - \sin \phi_{b2}). \quad (10b)$$

As noted from (10), when the IFOG is applied to small rotation exploration (namely, ϕ_s is close to zero), the value of excessive RIN approaches zero. More importantly, the RIN in the new scheme can be suppressed by adjusting the modulation depth difference. For the minimum configuration with conventional square-wave modulation, the demodulation process and the impact of RIN are provided in Appendix. The self-noise of conventional square-wave modulation is directly related to the demodulation factor μ , which is defined as the ratio between the difference and the sum of two states. As noted from (A3), both the numerator and denominator of μ are influenced by I_p^{RIN1} and I_p^{RIN2} . Since they are statistically independent, RIN significantly affects the demodulation process. Compared with the conventional scheme, cooperative four-state modulation can theoretically enhance the sensitivity of IFOG.

IV. MODULATION DEPTH OPTIMIZATION

As mentioned in Section II, there are four types of noise that affect the sensitivity of IFOG, namely RIN, TPN, shot noise, and Johnson–Nyquist noise. In the case of conventional square-wave modulation, the modulation depth significantly affects the impact of each type of noise on the demodulation results. In the principle of cooperative four-state modulation mentioned in Section III, the modulation depth not only affects the performance of the four types of noise but also influences the effectiveness of RIN suppression. To provide guidance for the selection of modulation depths in the new scheme, it is necessary to establish a noise model that matches the actual situation.

Numerical simulations are conducted, where all parameters remain consistent with subsequent experiments. The PMF coil length L is set to 520 m, coil diameter D to 0.098 m, and optical power I_p^{dc} to 0.42 mW (−3.8 dBm). Sagnac phase ϕ_s is set to 3.227×10^{-5} rad, corresponding to the phase of the projected Earth rotation rate at the location of Beijing. The sampling rate f_s is set to 6 MHz in both experiments and simulations to ensure compatibility with sensitivity and demodulation hardware performance. To ensure synchronization and reliability, the modulation frequency f_m is set to 200 kHz. To ensure the accuracy and effectiveness of the simulation, all noise parameters are obtained through fitting based on experimental results from conventional square-wave modulation. According to (3), RIN is modeled as a zero-mean Gaussian random process superimposed on the optical power. The level of RIN is determined by the SNR of the optical power, that is, $20\log_{10}(I_p^{dc}/I_p^{RIN})$. The SNR of optical power is determined by experimental results near a modulation depth of $\pi/2$ rad [29], [30], resulting in a final value of 58.2 dB. Shot noise and Johnson–Nyquist noise are equivalent to a Gaussian white noise superimposed on the detected optical intensity. Based on the existing RIN level and experimental results near a modulation depth of π rad [29], the standard deviation of noise is fit as 1.85×10^{-8} W. TPN is simplified as white noise [9] and its level is determined by the SNR of the Sagnac phase, that is, $20\log_{10}(\phi_s/\phi_{TPN})$. Using established noise parameters and the ultimate sensitivity from conventional square-wave experiments, the SNR of the Sagnac phase is determined to be 20.0 dB.

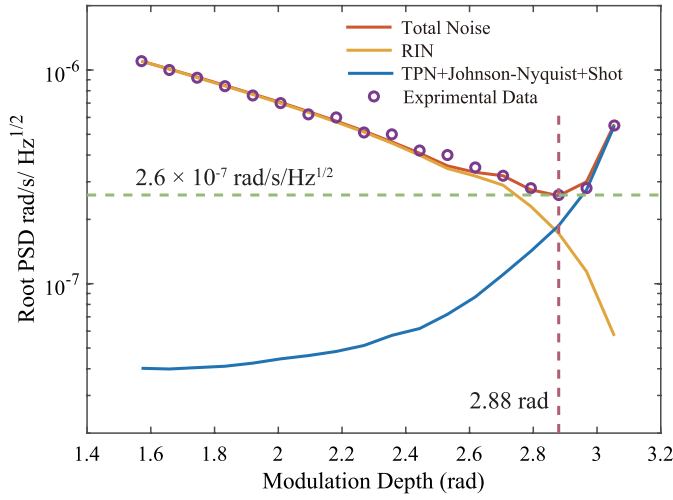


Fig. 3. Simulated noise level of conventional square-wave modulation. The conventional square-wave modulated IFOG achieves the sensitivity limit at a modulation depth of 2.88 rad. Below a modulation depth of 2.88 rad, RIN dominates as the primary noise source. Above a modulation depth of 2.88 rad, shot noise and Johnson–Nyquist noise become the main sources of noise.

The simulation and experimental results are depicted in Fig. 3. The simulation results closely match the experimental results, and the validity of the noise model is confirmed. Several conclusions can be drawn from it. The impact of RIN decreases gradually with increasing modulation depth, while the sum of other noises increases with it. When the modulation depth is below 2.88 rad, RIN becomes the primary source of noise. When the modulation depth is above 2.88 rad, shot noise and Johnson–Nyquist noise become the main sources of noise. Neither of the two cases achieve the highest sensitivity. In simulations and experiments, when the modulation depth reaches 2.88 rad, the conventional square-wave modulated IFOG achieves the highest sensitivity, the self-noise reaching 2.6×10^{-7} rad/s/Hz^{1/2}. In summary, the sensitivity performance of conventional square-wave modulation is highly correlated with the absolute value of the modulation depth. When external factors (such as reference voltage) change, the level of self-noise will also be altered.

In order to validate the effectiveness of the proposed scheme and select a set of appropriate modulation depths, numerical simulations are conducted on cooperative four-state modulation using the existing noise model. Notably, RIN in both PMF coils is consistent, while TPN, shot noise, and Johnson–Nyquist noise are statistically independent. The modulation depth ranges from $\pi/2$ rad to π rad and the results are depicted in Fig. 4.

Owing to the equitable status of ϕ_{b1} and ϕ_{b2} , the overall image exhibits axisymmetric characteristics centered around $\phi_{b1} = \phi_{b2}$. By selecting suitable modulation depths, significant reductions in the noise level of IFOG can be achieved. For a large difference in modulation depths, the RIN suppression is less effective, resulting in high noise levels, as shown by the prominent peaks on both sides in Fig. 4. When the modulation depths ϕ_{b1} and ϕ_{b2} are relatively close and at higher values, RIN is handled but Johnson–Nyquist noise and

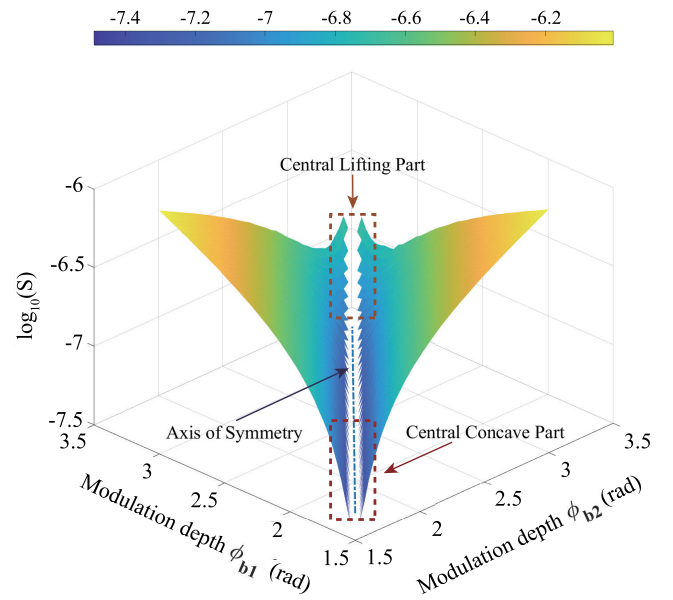


Fig. 4. Simulated noise performance of IFOG with cooperative square-wave modulation. S represents the root PSD of the noise with unit of rad/s/Hz^{1/2}. As ϕ_{b1} and ϕ_{b2} get closer, enhanced RIN suppression is observed. When both ϕ_{b1} and ϕ_{b2} are in the “central concave part,” the IFOG exhibits the optimal sensitivity.

shot noise become dominant, leading to secondary peaks in the “central lifting part” in Fig. 4. However, when ϕ_{b1} and ϕ_{b2} are relatively close and are at lower values, RIN is effectively suppressed, while the influence of Johnson–Nyquist noise and shot noise is well-handled. The phenomenon is manifested in the “central concave part” in Fig. 4. By adjusting ϕ_{b1} and ϕ_{b2} to the “central concave part” shown in Fig. 4, the highest sensitivity can be achieved. Due to the demodulation requirement that $\phi_{b1} \neq \phi_{b2}$, we ultimately control the difference between ϕ_{b1} and ϕ_{b2} to be 0.1 rad and their absolute values near $\pi/2$.

It also implies that in the “central concave part” shown in Fig. 4, the sensitivity of the IFOG is mainly related to the difference in modulation depths, reducing the dependence on the absolute values of them. Maintaining a constant modulation depth in an open-loop structure poses considerable engineering challenges. However, the modulation depth variation caused by external factors such as reference voltage drift exhibits a high common-mode component in the two minimum configurations. The modulation depth fluctuations in common mode do not affect the sensitivity in the new scheme. It is also an improvement over conventional scheme.

V. EXPERIMENTAL VERIFICATION

To validate the effectiveness of the cooperative four-state modulation scheme in improving sensitivity, a series of verification experiments are conducted. The experiments employ an ASE light source (BG-ASE-M1-C, Beogold Technology, Xiamen, China) with a central wavelength of 1550 nm and a spectral width of 35 nm, set to an output power of 13 dBm. Environmental temperature fluctuations introduce phase noise in IFOGs, known as the Shupe effect [31], [32]. To handle the Shupe effect, dual PMF coils are wound in quadrupole symmetric method [8], and temperature control measures are

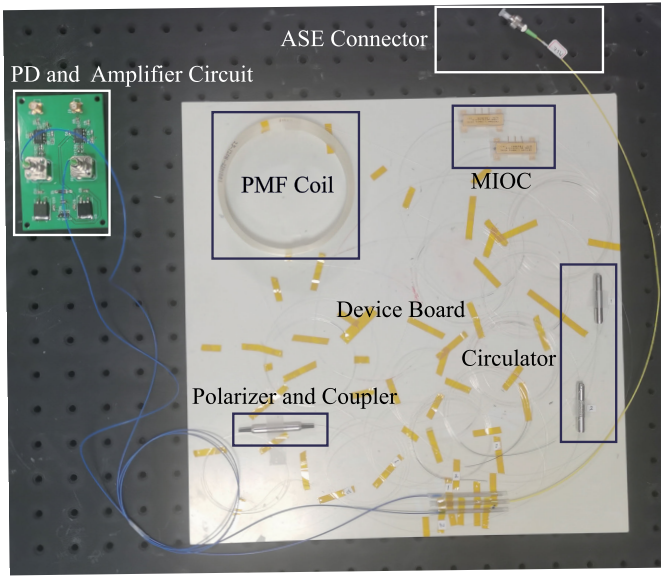


Fig. 5. Photograph of the optical and electric devices.

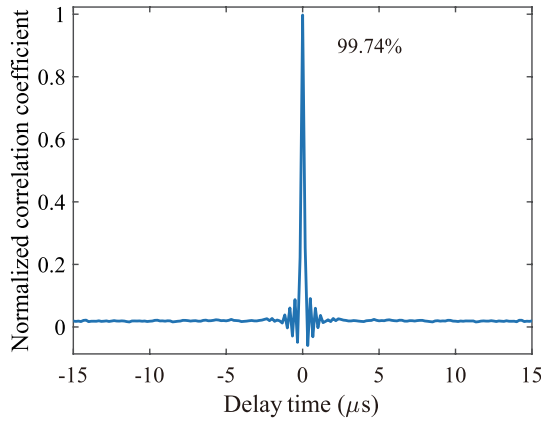


Fig. 6. Normalized correlation coefficient between the intensities of two minimum configurations.

implemented during experiments to ensure that phase errors caused by the Shupe effect can be neglected. Each PMF coil has a diameter of 0.098 m and a length of 520 m, with a fiber diameter of 80 μm . PD1 and PD2 (LSIPD-UL0.3-B-4JKFC, Minguang Technology, Beijing, China) are precisely calibrated to ensure consistency. The detection signal is collected by PXI-5922 (National Instruments, Austin, USA) at a sampling rate of 6 million samples per second (MSPS). Optical and electric devices are illustrated in Fig. 5.

A. Experimental Verification of RIN Consistency

The consistency of RIN in both PMFs is a necessary condition for noise suppression. We collect intensity signals in PD1 and PD2 with no modulation signal applied. The normalized correlation coefficient between them is calculated, and the result is plotted in Fig. 6. It can be observed that the noise correlation reaches 99.7%, which allows the two minimum configurations to serve as reference paths for each other. From another perspective, TPN, shot noise, and Johnson–Nyquist noise in both PMFs exhibit little correlation. Such a

remarkable level of correlation further substantiates the significant role played by RIN.

B. Experimental Verification of Sensitivity Enhancement

To assess the capability of cooperative four-state modulation in enhancing the sensitivity of an IFOG, we conduct a series of detection experiments targeting the Earth's rotation. In the experiment, the PMF coils are placed on horizontal ground, sensing the projection of the Earth's rotation in Beijing (9.67 deg/h, 39°59' N). All experiments are conducted in a stable laboratory environment.

According to the modulation depth selection principle presented in Section IV, we adjust the difference in modulation depths to 0.1 rad and set their absolute values in the vicinity of $\pi/2$. To further illustrate the enhanced sensitivity of this scheme, we perform conventional sinusoidal modulation and conventional square-wave modulation with minimum configuration in the same environment. To make the comparison more convincing, we set the modulation depth of sinusoidal modulation to 2.7 rad [33], and the modulation depth of square-wave modulation to 2.88 rad to achieve the lowest noise level of both schemes. Each scheme is tested for 1 h.

Allan variance analysis and root PSD are utilized to estimate the test results. Fig. 7(a) and (b) display the Allan standard deviation curves and root PSD, respectively. Table I summarizes the self-noise level, angle random walk (ARW), and bias instability (BI) for the three sets of data.

The self-noise level and ARW characterize the sensitivity of the IFOG, while BI characterizes its long-term stability. In the case of the IFOG employing sinusoidal modulation, the self-noise level in the root PSD is 1.1×10^{-6} rad/s/Hz^{1/2}. The ARW and BI in the Allan curve are 2.7×10^{-3} deg/h^{1/2} and 9.5×10^{-3} deg/h, respectively. For the IFOG utilizing the conventional square-wave modulation, the self-noise level in the root PSD is 2.6×10^{-7} rad/s/Hz^{1/2}. The ARW and BI in the Allan curve are 6.4×10^{-4} deg/h^{1/2} and 4.8×10^{-3} deg/h, respectively. However, the IFOG implementing the newly proposed scheme significantly reduces the self-noise level to 5.5×10^{-8} rad/s/Hz^{1/2}, exhibiting a remarkable reduction of 20.0-fold and 4.7-fold compared to the sinusoidal modulation and conventional square-wave modulation, respectively. Notably, the ARW in the Allan curve demonstrates a marked reduction of 18.0-fold and 4.3-fold, respectively, with a value of 1.5×10^{-4} deg/h^{1/2}. These findings indicate an enhancement in the sensitivity of an IFOG adopting this scheme. At the same time, the long-term stability of the IFOG is ensured. The BI in the Allan curve exhibits a reduction of 4.1-fold and 2.1-fold with a value of 2.3×10^{-3} deg/h compared to the reference groups.

The experimental results indicate the effectiveness of this scheme in RIN suppressing, significantly reducing the self-noise level and ARW in the IFOG.

VI. DISCUSSION

The source-sharing configuration plays a crucial role in RIN suppression. Unlike previous schemes that involve adding an optical reference arm in the optical path, this scheme enables

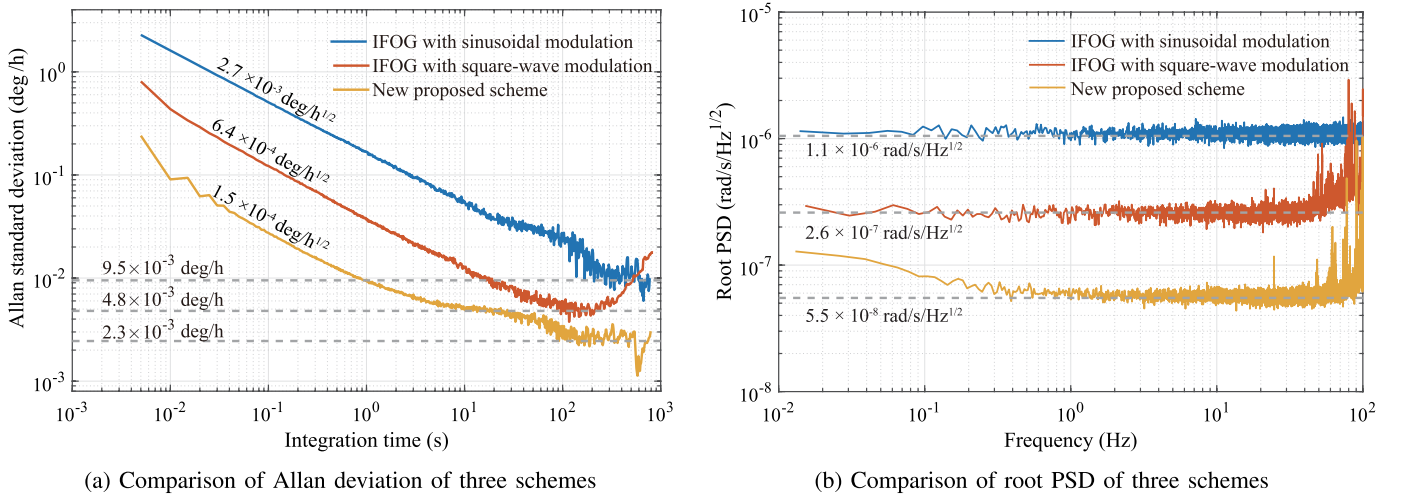


Fig. 7. Performance analysis of the cooperative four-state modulation scheme, with conventional sinusoidal modulation and conventional square-wave modulation as control groups. (a) Comparison of Allan deviation of three schemes. (b) Comparison of root PSD of three schemes.

TABLE I
PERFORMANCE COMPARISON OF THIS WORK AND WIDELY USED MODULATION SCHEME

Category	Self-noise (rad/s/Hz ^{1/2})	Improved factor	ARW (deg/h ^{1/2})	Improved factor	BI (deg/h)	Improved factor	Coil area (m ²)
Sinusoidal modulation	1.1×10^{-6}	1	2.7×10^{-3}	1	9.5×10^{-3}	1	12.74
Conventional square-wave modulation	2.6×10^{-7}	4.2	6.4×10^{-4}	4.2	4.8×10^{-3}	2.0	12.74
Cooperative four-state modulation	5.5×10^{-8}	20.0	1.5×10^{-4}	18.0	2.3×10^{-3}	4.1	12.74×2

TABLE II
PERFORMANCE COMPARISON OF THIS WORK AND HIGH-PERFORMANCE IFOGS

Category	Self-noise (rad/s/Hz ^{1/2})	ARW (deg/h ^{1/2})	BI (deg/h)	Coil area (m ²)
SRS-1001 [34]	$1.5 \times 10^{-7*}$	5.0×10^{-4}	6.0×10^{-4}	37.5
IXSea[FOG180] [34]	$6.4 \times 10^{-8*}$	2.2×10^{-4}	7.0×10^{-4}	67.5
IFOG-Based 3C Rotational Seismometer [7]	1.2×10^{-7}	2.7×10^{-4}	7.0×10^{-4}	145
Dual-source configuration [35]	$1.3 \times 10^{-7*}$	4.6×10^{-4}	4.6×10^{-4}	67
This work	5.5×10^{-8}	1.5×10^{-4}	2.3×10^{-3}	25.5

* Calculated from the Allan standard deviation curves.

two sets of parameter-matched gyroscopes to reference each other, providing a prerequisite for consistent RIN. Cooperative four-state modulation creates four states with strong RIN correlation. The modulation states implemented cooperatively serve as references to each other, effectively eliminating the influence of RIN. The noise model provides a basis for modulation depth selection. By choosing appropriate modulation depths, effective RIN suppression is achieved, while Johnson–Nyquist noise and shot noise are well-handled.

Table II presents some high-performance gyroscopes in the industrial and academic sectors. SRS-1001 [34], IXSea [FOG180] [34], and IFOG-based 3C rotational seismometer [7] are commercial instruments. The dual-source configuration [35] represents a new achievement in academia. It utilizes polarization multiplexing and two independent light sources to enhance performance of the IFOG. Our proposed scheme involves a smaller coil area while demonstrating lower level of self-noise and ARW. The comparison shows the remarkable capability of cooperative four-state modulation in noise suppression. Additionally, this scheme does not introduce

any external feedback system, avoiding sensitivity reduction caused by dead zone effects.

VII. CONCLUSION

We present the cooperative four-state modulation scheme based on a source-sharing configuration. The overall optical configuration consists of two identical minimum configuration with the same set of parameters, utilizing a common ASE light source and front polarizer to ensure RIN correlation. Cooperative four-state modulation effectively utilizes this strong correlation, allowing the cooperative modulation states to serve as references to each other and providing a solution for RIN suppression under square-wave modulation. The noise mechanism of the IFOG is extensively analyzed in this article, and a simulation model that aligns with test is established. The model is employed to predict the RIN suppression capability of the proposed scheme and provide modulation depth criteria.

The effectiveness of the scheme has been demonstrated through experiments. In comparison to the minimum configurations based on conventional sinusoidal modulation and

square-wave modulation, the scheme achieves a reduction in self-noise level by a factor of 20.0 and 4.7, respectively, as well as a decrease in ARW by a factor of 18.0 and 4.3, respectively. Given its remarkable effectiveness in enhancing sensitivity, the scheme can be applied to various sensing fields, especially in the rapidly advancing field of rotational seismology. Using this scheme to develop compact rotational seismometer can lead to space savings and improved detection capabilities for weak signals in such instruments.

APPENDIX

For the IFOG using conventional square-wave modulation, the two states of the modulation cycle are represented as follows:

$$I_1 = I_p + I_p \cos(\phi_s + \phi_b) \quad (\text{A1a})$$

$$I_2 = I_p + I_p \cos(\phi_s - \phi_b) \quad (\text{A1b})$$

where I_A and I_B are the signals at the phase biased states ϕ_b and $-\phi_b$. In the open-loop configuration, the Sagnac phase ϕ_s can be calculated by the following equation:

$$\phi_s = -\text{sign} \cdot \arccos \left[\sin^2 \phi_b (1 - \mu^2)^{\frac{1}{2}} - \frac{\mu^2 \cos \phi_b}{\mu^2 \cos^2 \phi_b + \sin^2 \phi_b} \right] \quad (\text{A2})$$

with

$$\text{sign} = \begin{cases} 1, & \mu \geq 0 \\ -1, & \mu < 0 \end{cases}, \quad \mu = \frac{I_1 - I_2}{I_1 + I_2}.$$

Taking the minimum configuration containing PMF A in Fig. 1 as an example, the excess RIN contained in the numerator and denominator of μ are represented as follows:

$$I_{A1}^{\text{RIN}} - I_{A2}^{\text{RIN}} = I_p^{\text{RIN1}} - I_p^{\text{RIN2}} + I_p^{\text{RIN1}} \cos(\phi_s + \phi_{b1}) - I_p^{\text{RIN2}} \cos(\phi_s - \phi_{b1}) \quad (\text{A3a})$$

$$I_{A1}^{\text{RIN}} + I_{A2}^{\text{RIN}} = I_p^{\text{RIN1}} + I_p^{\text{RIN2}} + I_p^{\text{RIN1}} \cos(\phi_s + \phi_{b1}) + I_p^{\text{RIN2}} \cos(\phi_s - \phi_{b1}). \quad (\text{A3b})$$

Due to the statistical independence of I_p^{RIN1} and I_p^{RIN2} , their impact on μ cannot cancel out. Consequently, RIN results in reduced sensitivity in the conventional square-wave modulation scheme.

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